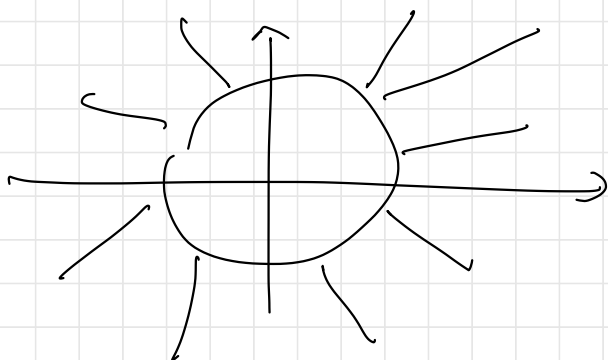


- Residuo all' infinito
- Funzioni armoniche
- Funzioni polidrome

Residuo all' infinito

Df. ∞ è una singolarità isolata per f
 se f è olomorfa in $U(\infty)$ ~~non~~
 (ovvero al di fuori di un disco centrato in 0)



$$|z| > R$$

$$\iff$$

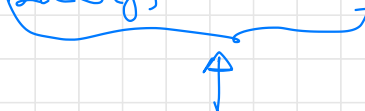
$$\left| \frac{1}{z} \right| < \frac{1}{R}$$

ovvero
 $\Leftrightarrow g(z) := f\left(\frac{1}{z}\right)$ ha una singolarità
 isolata in 0.

Teorema Se f è olomorfa su $\mathbb{C} \setminus \{ \text{un n° finito di punti} \}$,
 allora la somma di TUTTI i residui di f
 (compreso il $\text{Res}(f, \infty)$) è nulla

$$\left[\sum_{z_i \in S(f)} \text{Res}\left(f, \frac{z_i}{\infty}\right) + \text{Res}(f, \infty) = 0 \right]$$

$$\text{Def. } \text{Res}(f, \infty) := \text{Res} \left(-\frac{1}{z^2} f\left(\frac{1}{z}\right), 0 \right).$$

$$\sum_{z_i \in S(f)} \text{Res}(f, z_i) = -\text{Res}(f, \infty).$$


N.B. Nel Teo residui devo considerare

$$\sum_{z_i \in S(f)} \text{Res}(f, z_i) \text{Ind}(f, z_i)$$

Funzioni armoniche

Def. Sia $u: \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$
 $u(x_1, \dots, x_n)$

Si dice che u è ARMONICA in Ω se

$$\Delta u = 0 \quad \text{EQ. LAPLACE}$$



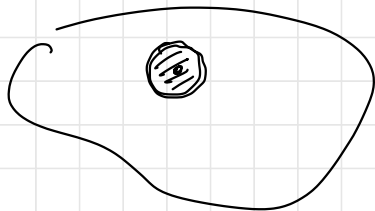
$$u_{x_1 x_1} + u_{x_2 x_2} + \dots + u_{x_n x_n} \leftarrow \text{laplaciano}$$

Esemp: $u(x_1, x_2) = a_1 x_1 + a_2 x_2 + b$

$(n=1 : u''=0 \quad u(x) = ax + b)$

$$u(x_1, x_2) = x_1^2 - x_2^2$$

$(n=2) \iff$ funzioni armoniche



n=2

Dom 1 f olomorfe $f = u + iv \Rightarrow$

u e v sono armoniche

$$(CR) \quad \begin{cases} u_x = v_y \\ u_y = -v_x \end{cases} \quad \begin{array}{l} \xrightarrow{D_x} \\ \xrightarrow{D_y} \end{array} \quad \begin{array}{l} u_{xx} = v_{yx} \\ u_{yy} = -v_{xy} \end{array}$$

$$\Delta u = u_{xx} + u_{yy} = \cancel{v_{yx}} - \cancel{v_{yx}} = 0$$

Dom 2 $u: \Omega \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ armonica,

Ω semp. connesso $\Rightarrow \exists v: \Omega \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$
armonica coniugata $\perp u$

tale che $f = u + iv$ olomorfe.

$$\omega = -u_y dx + u_x dy$$

$$\omega \text{ chiusa } (-u_y)_y = (u_x)_x$$

$$-u_{yy} = u_{xx}$$

$$u_{xx} + u_{yy} = 0.$$

Funzioni polidrome

$$z = |z| e^{i \operatorname{Arg} z}$$

$$\operatorname{Arg} z = \left\{ \underset{\substack{\uparrow \\ \text{arg. principale}}}{\arg z} + 2k\pi : k \in \mathbb{Z} \right\}$$

$\in [0, 2\pi)$

$$\begin{aligned} \sqrt[n]{z} &= \left\{ \sqrt[n]{|z|} \cdot e^{i \frac{\operatorname{Arg} z}{n}} \right\} \leftarrow n \text{ elementi distinti} \\ &= \left\{ \sqrt[n]{|z|} e^{i \left(\frac{\arg z}{n} + \frac{2k\pi}{n} \right)} : k=0, 1, \dots, n-1 \right\} \end{aligned}$$

$$\log z = \left\{ \log |z| + i \operatorname{Arg} z \right\} \leftarrow \infty \text{ elementi distinti}$$

$$\begin{aligned} w = \log |z| + i \operatorname{Arg} z &\Rightarrow e^w = e^{\log |z| + i \operatorname{Arg} z} = e^{\log |z|} \cdot e^{i \operatorname{Arg} z} \\ &= |z| \cdot e^{i \operatorname{Arg} z} = z \end{aligned}$$

Per definire $\sqrt[n]{z}$ come funzione

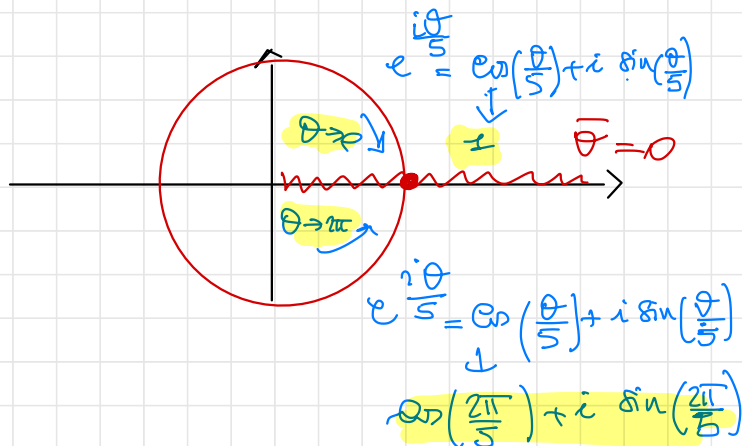
$$\mathbb{C} \ni z \mapsto \sqrt[n]{|z|} \cdot e^{i \frac{\operatorname{Arg} z}{n}} \quad \text{con}$$

$$\operatorname{Arg} z \in [0, 2\pi)$$

$$\operatorname{Arg} z \in [\bar{\theta}, \bar{\theta} + 2\pi)$$

BRANCA DELLA RADICE n -esima.

Ques Una branca di $\sqrt[n]{z}$ non è continua su $\mathbb{C}$.



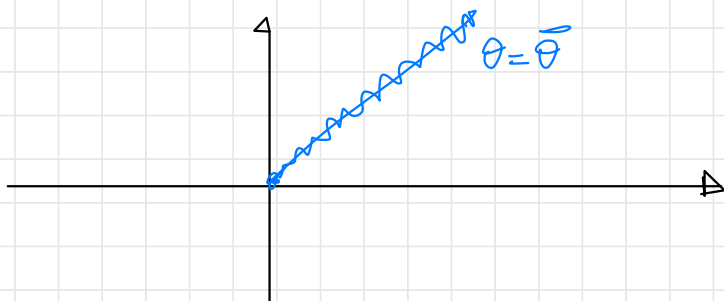
$$n = 5$$

$$\bar{\theta} = 0$$

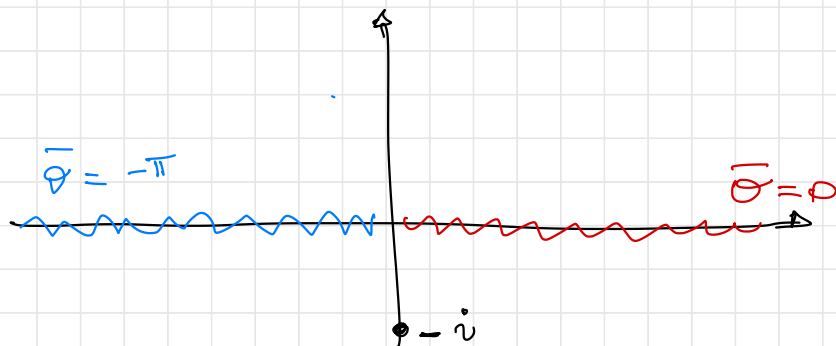
$$e^{i\frac{\theta}{5}} \text{ con}$$

$$\theta = [0, 2\pi]$$

è continua su $\mathbb{C} \setminus \{\theta = \bar{\theta}\}$



$$\theta \in [\bar{\theta}, \bar{\theta} + 2\pi]$$



$$f_1(z) = |z|^{1/5} e^{i \frac{\text{Arg } z}{5}} \quad \text{on } \text{Arg } z \in [0, 2\pi)$$

$$f_2(z) = |z|^{1/5} e^{i \frac{\text{Arg } z}{5}} \quad \text{on } \text{Arg } z \in [-\pi, \pi)$$

$$f_1(-i) = e^{i \frac{3\pi}{10}} = \cos\left(\frac{3\pi}{10}\right) + i \sin\left(\frac{3\pi}{10}\right)$$

$$\# f_2(-i) = e^{i \left(-\frac{\pi}{10}\right)} = \cos\left(-\frac{\pi}{10}\right) + i \sin\left(-\frac{\pi}{10}\right)$$

ANALISI FUNZIONALE

Def. Uno **SPAZIO VETTORIALE** su $\mathbb{R}$

è un insieme V su cui sono definite

$$\begin{cases} \text{Somma} & + : V \times V \longrightarrow V \\ \text{Prodotto per scalare} & \cdot : V \times \mathbb{R} \longrightarrow V \end{cases}$$

che godono delle proprietà:

$$\begin{array}{l} \bullet u + v = v + u \\ \bullet u + (v + w) = (u + v) + w \\ \bullet u + 0 = u \\ \bullet u + (-u) = 0 \end{array} \quad \left| \quad \begin{array}{l} \bullet (ts)u = t(su) \\ \bullet t(u+v) = tu + tv \\ \bullet (t+s)u = tu + su \\ \bullet 1 \cdot u = u \end{array} \right.$$

Esempi:

① $V = \mathbb{R}^n \leftarrow$ spazio euclideo (n -dim)

$$\begin{array}{l} \underline{x} = (x_1, \dots, x_n) \\ \underline{y} = (y_1, \dots, y_n) \end{array} \quad \begin{cases} \underline{x} + \underline{y} = (x_1 + y_1, \dots, x_n + y_n) \\ t\underline{x} = (tx_1, \dots, tx_n) \end{cases}$$

Dimensione finita (n).

BASE CANONICA

$$\begin{array}{l} \underline{e}_1 = (1, 0, \dots, 0) \\ \underline{e}_2 = (0, 1, \dots, 0) \\ \vdots \\ \underline{e}_n = (0, \dots, 1) \end{array}$$

② $V = \mathbb{R}[x] =$ polinomi nell'indeterminata x
e coeff in $\mathbb{R}$

$$\begin{cases} (p+q)(x) = p(x) + q(x) \\ (tp)(x) = t \cdot p(x) \end{cases}$$

↑
somma tra polinomi

↑
mult di un polinomio per uno scalare

dim. infinita

$$\{x^n\}_{n \in \mathbb{N}} = \{1, x, x^2, x^3, \dots\}$$

$$a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots + a_n x^n = 0$$

↑

$$\Rightarrow a_i = 0$$

③ $V = \mathcal{C}^0([a, b]) = \{f: [a, b] \rightarrow \mathbb{R} \text{ continue}\}$

$$\begin{cases} (f+g)(x) = f(x) + g(x) \\ (tf)(x) = t \cdot f(x) \end{cases}$$

↑
somma in $\mathbb{R}$

↑
prodotto in $\mathbb{R}$

$$V = \mathcal{C}^k([a, b])$$

Def. Sia V uno sp. vettoriale su $\mathbb{R}$

Una **NORMA** su V è una funzione

$$\| \cdot \| : V \rightarrow \mathbb{R}_+ = [0, +\infty) \text{ tale che}$$

1) $\|v\| > 0 \quad \forall v \in V \setminus \{0\}$ **POSITIVITÀ**

2) $\|tv\| = |t| \|v\| \quad \forall t \in \mathbb{R} \quad \forall v \in V$ **OMOGENEITÀ**

3) $\|u+v\| \leq \|u\| + \|v\| \quad \forall u, v \in V$ **TRIANGOLO**

Conseguenze

4) $\|0\| = 0$ Prendere $t=0$ Nella 2)
 $\|0\| = 0$

5) $|\|u\| - \|v\|| \leq \|u-v\| \quad \forall u, v \in V$

$$u = (u-v) + v$$

3) $\Rightarrow \|u\| = \|(u-v) + v\| \stackrel{3)}{\leq} \|u-v\| + \|v\|$

$$\Rightarrow \|u-v\| \geq \|u\| - \|v\|$$

stessi passaggi con $u \leftarrow v$

$$\Rightarrow \|u-v\| \geq \|v\| - \|u\|$$

Exemp:

① $V = \mathbb{R}^n$ $x = (x_1, \dots, x_n)$

• $\|x\|_2 = \sqrt{x_1^2 + \dots + x_n^2} = \left(\sum_{i=1}^n x_i^2 \right)^{1/2}$
norme euclidéenne

$n=1$ $\|x\|_2 = \sqrt{x^2} = |x|$

• $\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$ $p \in [1, +\infty)$

$p=1$ $\|x\|_1 = \sum_{i=1}^n |x_i|$

• $\|x\|_\infty = \max_{i=1, \dots, n} |x_i|$

$\|x\|_\infty = \lim_{p \rightarrow +\infty} \|x\|_p$

exercice!

$x = (x_1, x_2)$

$\lim_{p \rightarrow +\infty} (|x_1|^p + |x_2|^p)^{1/p} = \max(|x_1|, |x_2|)$

- $V = C^0([a, b])$

- $\|f\|_{\infty} := \max_{x \in [a, b]} |f(x)|$

- $\|f\|_1 := \int_a^b |f(x)|$

Def. $(V, \|\cdot\|)$ sp. vett. normato

Def. Sia $(V, \|\cdot\|)$ sp. vett. normato

Allora $d(u, v) := \|u - v\|$ definisce una

DISTANZA su V , ovvero $d: V \times V \rightarrow \mathbb{R}$ tale che

- 1) $d(u, v) \geq 0$, con $= \Leftrightarrow u = v$ POSITIVITA'
- 2) $d(u, v) = d(v, u)$ SIMMETRIA
- 3) $d(u, v) \leq d(u, w) + d(w, v)$ TRIANGOLOARE.

Def. (V, d) sp. metrico.

Def. Sia $(V, \|\cdot\|)$ normato ($\Rightarrow$ metrico)

Fissato $v_0 \in V$, $r > 0$

$$B_r(v_0) := \{v \in V: \|v - v_0\| < r\}$$

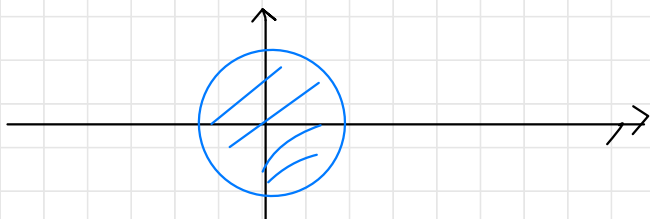
$\uparrow$
 $d(v, v_0)$

Sfera di centro v_0
e raggio r

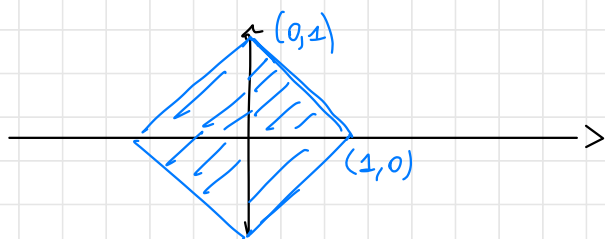
Exemp:

1) $V = \mathbb{R}^2$ $B_1(0)$

• $\|\cdot\|_2$ $B_1(0) = \{x \in \mathbb{R}^2 : \sqrt{x_1^2 + x_2^2} < 1\}$

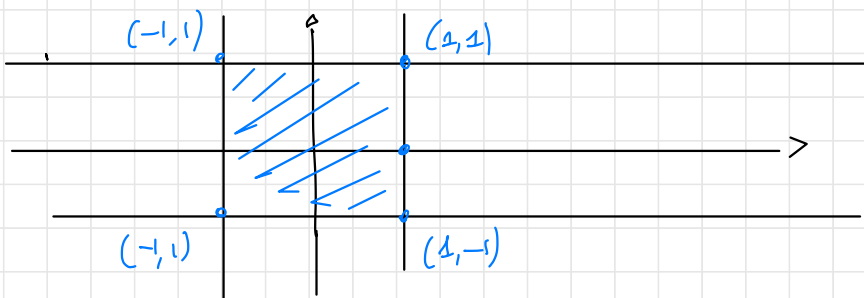


• $\|\cdot\|_1$ $B_1(0) = \{x \in \mathbb{R}^2 : \overbrace{|x_1| + |x_2|}^{\|x\|_1} < 1\}$

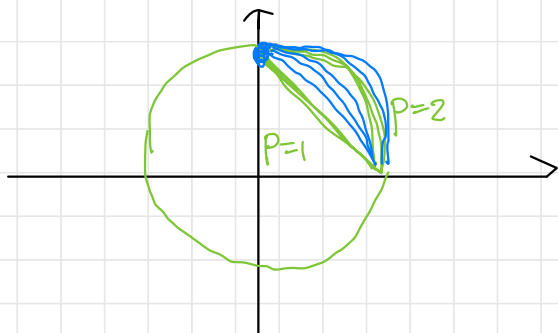


$$x_1 + x_2 < 1$$

• $\|\cdot\|_\infty$ $B_1(0) = \{x \in \mathbb{R}^2 : \max\{|x_1|, |x_2|\} < 1\}$



• $\|\cdot\|_p$ $B_1(0) = \{x \in \mathbb{R}^2 : (|x_1|^p + |x_2|^p)^{1/p} < 1\}$



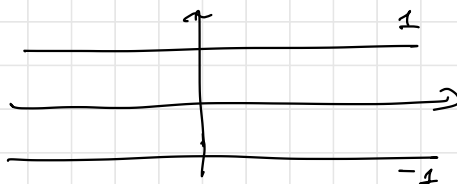
$$\begin{aligned} |x_1|^p + |x_2|^p &< 1 \\ |x_2|^p &< 1 - |x_1|^p \\ |x_2| &< \sqrt[p]{1 - |x_1|^p} \end{aligned}$$

$p=1 \rightarrow 1 - |x_1|$

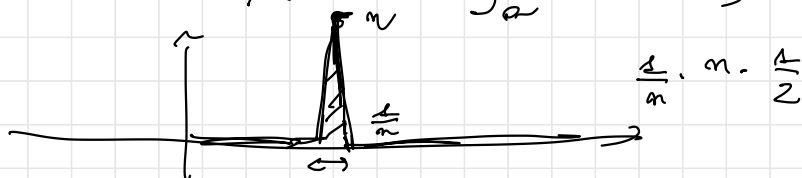
$p=2 \rightarrow \sqrt{1 - |x_1|^2}$

• $V = \mathcal{C}^0([a, b])$

$\|\cdot\|_\infty$ $B_1(0) = \{f \in V : \max_{[a, b]} |f| < 1\}$



$\|\cdot\|_1$ $B_1(0) = \{f \in V : \int_a^b |f| < 1\}$



Differenze tra sp. vet. di dim finita / infinito

① In dim finita tutte le norme su V sono "equivalenti".

In dim infinite FALSO

Def. Sia $\{v_n\} \subseteq (V, \|\cdot\|)$ una successione in uno sp. vet. normato. Si dice che

$$\lim_{n \rightarrow +\infty} v_n = \underset{\substack{V \\ \uparrow \\ V}}{v} \quad \text{se} \quad \lim_{n \rightarrow +\infty} \underbrace{\|v_n - v\|}_{d(v_n, v)} = 0$$

Def. Sia V uno sp. vett su $\mathbb{R}$ e siano

$\nearrow \|\cdot\|$

due norme su V .

$\searrow \|\cdot\|$

Qs due norme sono equivalenti se:

1) $\exists C > 0: \quad \forall v \in V \quad \|v\| \leq C \|\cdot\|$

2) $\exists C' > 0: \quad \forall v \in V \quad \|\cdot\| \leq C' \|v\|$

Se $\|\cdot\|$ e $\|\cdot\|$ sono equivalenti:

Se $v_m \rightarrow v$ in $\|\cdot\|$, allora $v_m \rightarrow v$ in $\|\cdot\|$

e viceversa

$$C' \cdot \|v_m - v\| \rightarrow 0$$

$$\|v_m - v\|$$

Esempio: $V = \mathbb{R}^2$, $\|\cdot\|_1$ e $\|\cdot\|_\infty$

sono equivalenti:

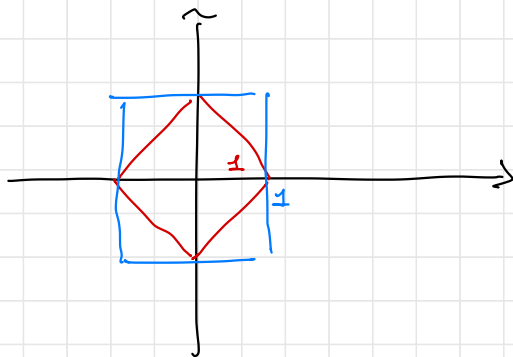
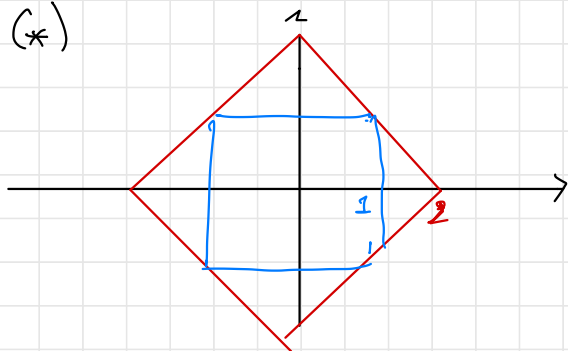
$$1) |x_1| + |x_2| \leq \underset{\substack{\uparrow \\ 2}}{C} \max\{|x_1|, |x_2|\} \quad \forall x \in \mathbb{R}^2$$

$$2) \max\{|x_1|, |x_2|\} \leq \underset{\substack{\uparrow \\ 1}}{C'} (|x_1| + |x_2|) \quad \forall x \in \mathbb{R}^2$$

$$1) \|\cdot\|_1 \leq 2 \|\cdot\|_\infty \iff B_1^{\infty}(0) \subseteq B_2^1(0) \quad (*)$$

$$2) \|\cdot\|_\infty \leq \|\cdot\|_1 \iff B_1^1(0) \subseteq B_2^{\infty}(0) \quad (**)$$

(*)



Teorema: $\dim V < +\infty \Rightarrow$ tutte le norme su V sono equivalenti.

FALSO in dim infinita

Controesempio

$$V = C^0([a, b])$$

$$\|f\|_\infty = \max_{x \in [a, b]} |f(x)|$$

$$\|f\|_1 = \int_a^b |f|.$$

NON equivalenti

$$\|f\|_1 = \int_a^b |f| \leq (b-a) \max |f| = \underbrace{(b-a)}_C \|f\|_\infty$$

$$\nexists C : \|f\|_\infty \leq C \|f\|_1$$

$$\exists f_n : \frac{\|f_n\|_\infty}{\|f_n\|_1} \longrightarrow +\infty.$$

